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Project 1: Student Management System :

```

project1.py > ...
1  # Student Management System (Simple Version)
2
3  students = {} # Dictionary to store student records
4
5  # Function to calculate grade based on percentage
6  def calculate_grade(percentage):
7      if percentage >= 90:
8          return "A+"
9      elif percentage >= 75:
10         return "A"
11     elif percentage >= 60:
12         return "B"
13     elif percentage >= 40:
14         return "C"
15     else:
16         return "Fail"
17
18 # Function to add a student
19 def add_student():
20     try:
21         roll = int(input("Enter Roll Number: "))
22         if roll in students:
23             print(" Roll number already exists!")
24             return
25         name = input("Enter Name: ")
26         marks = []
27         for i in range(1, 6):
28             mark = float(input(f"Enter marks for Subject {i}: "))
29             marks.append(mark)
30         percentage = sum(marks) / 5
31         grade = calculate_grade(percentage)
32         students[roll] = {"name": name, "marks": marks, "percentage": percentage, "grade": grade}
33         print(" Student added successfully!")
34     except ValueError:
35         print(" Invalid input! Please enter numbers correctly.")
36

```

```

37 # Function to view all students
38 def view_all():
39     if not students:
40         print("No records found!")
41     else:
42         for roll, details in students.items():
43             print(f"Roll: {roll}, Name: {details['name']}, %: {details['percentage']:.2f}, Gra
44
45 # Function to search student
46 def search_student():
47     roll = int(input("Enter Roll Number to search: "))
48     if roll in students:
49         print("Student Found:", students[roll])
50     else:
51         print(" Student not found!")
52
53 # Function to update student marks
54 def update_student():
55     roll = int(input("Enter Roll Number to update: "))
56     if roll in students:
57         new_marks = []
58         for i in range(1, 6):
59             mark = float(input(f"Enter new marks for Subject {i}: "))
60             (variable) percentage: float
61         percentage = sum(new_marks) / 5
62         grade = calculate_grade(percentage)
63         students[roll]["marks"] = new_marks
64         students[roll]["percentage"] = percentage
65         students[roll]["grade"] = grade
66         print(" Marks updated successfully!")
67     else:
68         print(" Student not found!")
69

```

```

70 # Function to delete student
71 def delete_student():
72     roll = int(input("Enter Roll Number to delete: "))
73     if roll in students:
74         del students[roll]
75         print(" Student deleted successfully!")
76     else:
77         print(" Student not found!")
78
79 # Main Menu Loop
80 while True:
81     print("\n===== STUDENT MANAGEMENT SYSTEM =====")
82     print("1. Add Student")
83     print("2. View All")
84     print("3. Search")
85     print("4. Update")
86     print("5. Delete")
87     print("6. Exit")
88
89     choice = input("Enter choice (1-6): ")
90
91     if choice == "1":
92         add_student()
93     elif choice == "2":
94         view_all()
95     elif choice == "3":
96         search_student()
97     elif choice == "4":
98         update_student()
99     elif choice == "5":
100        delete_student()
101     elif choice == "6":
102        print("Exiting... Goodbye!")
103        break
104     else:
105        print(" Invalid choice! Please enter between 1-6.")

```

Output :

```

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All
3. Search
4. Update
5. Delete
6. Exit
Enter choice (1-6): 1
Enter Roll Number: 23
Enter Name: ABC
Enter marks for Subject 1: 88
Enter marks for Subject 2: 77
Enter marks for Subject 3: 99
Enter marks for Subject 4: 90
Enter marks for Subject 5: 76
  Student added successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. View All
3. Search
4. Update
5. Delete
6. Exit
Enter choice (1-6): 2
Roll: 23, Name: ABC, %: 86.00, Grade: A

```

Project 2: Library Management System :

```
project2.py > ...
1  # Library Management System (Simple Version)
2  |
3  import datetime
4  |
5  library = {} # Dictionary to store books (ISBN → details)
6  |
7  # Function to add a book
8  def add_book():
9      isbn = input("Enter ISBN: ")
10     if isbn in library:
11         print(" Book with this ISBN already exists!")
12         return
13     title = input("Enter Title: ")
14     author = input("Enter Author: ")
15     library[isbn] = {
16         "title": title,
17         "author": author,
18         "available": True,
19         "borrower": None,
20         "issue_date": None
21     }
22     print(" Book added successfully!")
23 |
24 # Function to issue a book
25 def issue_book():
26     isbn = input("Enter ISBN to issue: ")
27     if isbn in library:
28         if library[isbn]["available"]:
29             borrower = input("Enter Borrower Name: ")
30             student_id = input("Enter Student ID: ")
31             library[isbn]["available"] = False
32             library[isbn]["borrower"] = f"{borrower} ({student_id})"
33             library[isbn]["issue_date"] = datetime.date.today()
34             due_date = library[isbn]["issue_date"] + datetime.timedelta(days=7)
35             print(" Book issued successfully!")
36             print(f"Title: {library[isbn]['title']}, Due Date: {due_date}")
37 |
38         else:
39             print(" Book already issued!")
40     else:
41         print(" Book not found!")
42 |
43 # Function to return a book
44 def return_book():
45     isbn = input("Enter ISBN to return: ")
46     if isbn in library:
47         if not library[isbn]["available"]:
48             library[isbn]["available"] = True
49             library[isbn]["borrower"] = None
50             library[isbn]["issue_date"] = None
51             print(" Book returned successfully!")
52         else:
53             print(" Book was not issued!")
54     else:
55         print("Book not found!")
56 |
57 # Function to search book by title or author
58 def search_book():
59     keyword = input("Enter keyword (title/author): ").lower()
60     found = False
61     for isbn, details in library.items():
62         if keyword in details["title"].lower() or keyword in details["author"].lower():
63             print(f"ISBN: {isbn}, Title: {details['title']}, Author: {details['author']}, Available: {details['available']}")
64             found = True
65     if not found:
66         print(" No matching book found!")
```

```

67 # Function to view all books
68 def view_catalog():
69     if not library:
70         print("No books in catalog!")
71     else:
72         for isbn, details in library.items():
73             status = "Available" if details["available"] else f"Issued to {details['borrower']}"
74             print(f"ISBN: {isbn}, Title: {details['title']}, Author: {details['author']}, Status: {status}")
75
76 # Main Menu Loop
77 while True:
78     print("\n===== LIBRARY MANAGEMENT SYSTEM =====")
79     print("1. Add Book")
80     print("2. Issue Book")
81     print("3. Return Book")
82     print("4. Search Book")
83     print("5. View Catalog")
84     print("6. Exit")
85
86     choice = input("Enter choice (1-6): ")
87
88     if choice == "1":
89         add_book()
90     elif choice == "2":
91         issue_book()
92     elif choice == "3":
93         return_book()
94     elif choice == "4":
95         search_book()
96     elif choice == "5":
97         view_catalog()
98     elif choice == "6":
99         print("Exiting... Goodbye!")
100        break
101    else:
102        print("Invalid choice! Please enter between 1-6.")
103

```

Output :

```

ers/Sarvesh/OneDrive/Desktop/code/project2.py

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Issue Book
3. Return Book
4. Search Book
5. View Catalog
6. Exit
Enter choice (1-6): 1
Enter ISBN: 12345
Enter Title: Python Basic
Enter Author: sarvesh
Book added successfully!

===== LIBRARY MANAGEMENT SYSTEM =====
1. Add Book
2. Issue Book
3. Return Book
4. Search Book
5. View Catalog
6. Exit
Enter choice (1-6): 4
Enter keyword (title/author): sarvesh
ISBN: 12345, Title: Python Basic, Author: sarvesh, Available: True

```

Project 3: Personal Expense Tracker :

```
project3.py > add_expense
1  # Personal Expense Tracker (Simple Version)
2
3  expenses = [] # List to store expense records
4  budget = float(input("Enter your monthly budget: "))
5
6  # Function to add expense
7  def add_expense():
8      try:
9          desc = input("Enter description: ")
10         category = input("Enter category (Food/Travel/Bills/Entertainment/Other): ")
11         amount = float(input("Enter amount: "))
12         if amount <= 0:
13             print(" Amount must be positive!")
14             return
15         date = input("Enter date (DD-MM-YYYY): ")
16         expenses.append({"desc": desc, "category": category, "amount": amount, "date": date})
17         print(" Expense added successfully!")
18     except ValueError:
19         print(" Invalid amount! Please enter a number.")
20
21 # Function to view all expenses
22 def view_expenses():
23     if not expenses:
24         print("No expenses recorded!")
25     else:
26         print("\n===== ALL EXPENSES =====")
27         for i, exp in enumerate(expenses, start=1):
28             print(f"{i}. {exp['desc']} | {exp['category']} | Rs.{exp['amount']} | {exp['date']}")
29
```

```
project3.py > add_expense
30 # Function to show category summary
31 def category_summary():
32     summary = {}
33     for exp in expenses:
34         summary[exp["category"]] = summary.get(exp["category"], 0) + exp["amount"]
35     print("\n===== CATEGORY SUMMARY =====")
36     for cat, total in summary.items():
37         print(f"{cat}: Rs.{total}")
38
39 # Function to show budget report
40 def budget_report():
41     total = sum(exp["amount"] for exp in expenses)
42     used_percent = (total / budget) * 100
43     remaining = budget - total
44     print("\n===== BUDGET REPORT =====")
45     print(f"Total Spent : Rs.{total:.2f}")
46     print(f"Budget Limit : Rs.{budget:.2f}")
47     print(f"Remaining : Rs.{remaining:.2f}")
48     print(f"Used : {used_percent:.2f}%")
49     if used_percent >= 100:
50         print("⚠ WARNING: Budget exceeded!")
51     elif used_percent >= 80:
52         print("⚠ WARNING: You have used over 80% of your budget!")
53
54 # Find top category
55 if expenses:
56     summary = {}
57     for exp in expenses:
58         summary[exp["category"]] = summary.get(exp["category"], 0) + exp["amount"]
59     top_cat = max(summary, key=summary.get)
60     print(f"Top Category : {top_cat} (Rs.{summary[top_cat]:.2f})")
61
```

```

61
62 # Main Menu Loop
63 while True:
64     print("\n===== PERSONAL EXPENSE TRACKER =====")
65     print("1. Add Expense")
66     print("2. View All")
67     print("3. Category Summary")
68     print("4. Budget Report")
69     print("5. Exit")
70
71     choice = input("Enter choice (1-5): ")
72
73     if choice == "1":
74         add_expense()
75     elif choice == "2":
76         view_expenses()
77     elif choice == "3":
78         category_summary()
79     elif choice == "4":
80         budget_report()
81     elif choice == "5":
82         print("Exiting... Goodbye!")
83         break
84     else:
85         print(" Invalid choice! Please enter between 1-5.")
86

```

Output :

```

ers/Sarvesh/OneDrive/Desktop/code/project3.py
Enter your monthly budget: 50000

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All
3. Category Summary
4. Budget Report
5. Exit
Enter choice (1-5): 1
Enter description: Pizza
Enter category (Food/Travel/Bills/Entertainment/Other): Food
Enter amount: 1000
Enter date (DD-MM-YYYY): 02/12/2025
Expense added successfully!

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All
3. Category Summary
4. Budget Report
5. Exit
Enter choice (1-5): 2

===== ALL EXPENSES =====
1. Pizza | Food | Rs.1000.0 | 02/12/2025

```

Project 4: Quiz and Examination System :

```
project4.py > ...
1  # Quiz and Examination System (General Knowledge Version)
2
3  # Question bank stored as tuples (question, A, B, C, D, correct_option)
4  questions = [
5      ("What is the National Animal of India?", "Tiger", "Lion", "Elephant", "Leopard", "A"),
6      ("What is the National Bird of India?", "Peacock", "Sparrow", "Parrot", "Crow", "A"),
7      ("What is the National Anthem of India?", "Vande Mataram", "Jana Gana Mana", "Saare Jahan", "Vande M", "A"),
8      ("What is the National Song of India?", "Saare Jahan Se Achha", "Jana Gana Mana", "Vande M", "A"),
9      ("What is the National Flower of India?", "Lotus", "Rose", "Sunflower", "Jasmine", "A"),
10     ("What is the National Fruit of India?", "Apple", "Banana", "Mango", "Orange", "C"),
11     ("What is the National Tree of India?", "Neem", "Banyan", "Peepal", "Ashoka", "B"),
12     ("What is the National River of India?", "Yamuna", "Ganga", "Brahmaputra", "Godavari", "B"),
13     ("What is the National Currency of India?", "Dollar", "Rupee", "Pound", "Euro", "B"),
14     ("What is the National Sport of India?", "Cricket", "Hockey", "Kabaddi", "Football", "B")
15 ]
16
17 # Function to calculate grade
18 def calculate_grade(percent):
19     if percent >= 90:
20         return "A+"
21     elif percent >= 75:
22         return "A"
23     elif percent >= 60:
24         return "B"
25     elif percent >= 50:
26         return "C"
27     else:
28         return "Fail"
29
30 # Main quiz function
31 def start_quiz():
32     name = input("Enter your name: ")
33     roll = input("Enter your roll number: ")
34     score = 0
35     wrong_answers = []
36
37     print(f"\nWelcome {name} (Roll: {roll}) to the General Knowledge Quiz!\n")
38
39     for i, q in enumerate(questions, start=1):
40         print(f"Q{i}: {q[0]}")
41         print(f"A. {q[1]} B. {q[2]} C. {q[3]} D. {q[4]}")
42         ans = input("Your Answer (A/B/C/D): ").strip().upper()
43
44         if ans == q[5]:
45             print("✓ Correct!\n")
46             score += 1
47         else:
48             print("X Wrong!\n")
49             wrong_answers.append((q[0], ans, q[5]))
50
51     total = len(questions)
52     percent = (score / total) * 100
53     grade = calculate_grade(percent)
54
55     print("\n===== RESULT REPORT =====")
56     print(f"Name      : {name}")
57     print(f"Roll No   : {roll}")
58     print(f"Score     : {score} / {total}")
59     print(f"Percent   : {percent:.2f}%")
60     print(f"Grade     : {grade}")
61     print("Result    : PASS" if percent >= 50 else "Result : FAIL")
62
63     if wrong_answers:
64         print("\n===== WRONG ANSWERS =====")
65         for q, given, correct in wrong_answers:
66             print(f"Q: {q}")
67             print(f"Your Answer: {given}, Correct Answer: {correct}\n")
68
69     # Run the quiz
70     start_quiz()
71
```


Output :

```
ers/sarvesh/OneDrive/Desktop/Code/project4.py
Enter your name: ABC
Enter your roll number: 23

Welcome ABC (Roll: 23) to the General Knowledge Quiz!

Q1: What is the National Animal of India?
A. Tiger B. Lion C. Elephant D. Leopard
Your Answer (A/B/C/D): a
✓ Correct!

Q2: What is the National Bird of India?
A. Peacock B. Sparrow C. Parrot D. Crow
Your Answer (A/B/C/D): a
✓ Correct!

Q3: What is the National Anthem of India?
A. Vande Mataram B. Jana Gana Mana C. Saare Jahan Se Achha D. Maa Tujhe Salaam
Your Answer (A/B/C/D): b
✓ Correct!

Q4: What is the National Song of India?
A. Saare Jahan Se Achha B. Jana Gana Mana C. Vande Mataram D. Ae Mere Watan Ke Logon
Your Answer (A/B/C/D): a
X Wrong!

Q5: What is the National Flower of India?
A. Lotus B. Rose C. Sunflower D. Jasmine
Your Answer (A/B/C/D): █
```

Project 5: Inventory Management System :

```
1  # Simple Inventory Management System
2
3  inventory = {} # Dictionary to store products
4
5  # Add Product
6  def add_product():
7      pid = input("Enter Product ID: ")
8      if pid in inventory:
9          print("| Product ID already exists!")
10         return
11     name = input("Enter Product Name: ")
12     price = float(input("Enter Price: "))
13     qty = int(input("Enter Quantity: "))
14     reorder = int(input("Enter Reorder Level: "))
15     inventory[pid] = {"name": name, "price": price, "qty": qty, "reorder": reorder}
16     print(" Product added successfully!")
17
18 # Stock In
19 def stock_in():
20     pid = input("Enter Product ID: ")
21     if pid in inventory:
22         qty = int(input("Enter quantity to add: "))
23         inventory[pid]["qty"] += qty
24         print(" Stock updated!")
25     else:
26         print(" Product not found!")
27
28 # Stock Out
29 def stock_out():
30     pid = input("Enter Product ID: ")
31     if pid in inventory:
32         qty = int(input("Enter quantity to remove: "))
33         if qty <= inventory[pid]["qty"]:
34             inventory[pid]["qty"] -= qty
35             print(" Stock deducted!")
36         else:
37             print(" Not enough stock!")
38     else:
39         print(" Product not found!")
40
41 # View Inventory
42 def view_inventory():
43     print("\n===== INVENTORY =====")
44     for pid, details in inventory.items():
45         value = details["price"] * details["qty"]
46         print(f"{pid} | {details['name']} | Rs.{details['price']} | Qty:{details['qty']} | Value: Rs.{value}")
47
48 # Low Stock Alert
49 def low_stock_alert():
50     print("\n===== LOW STOCK ALERT =====")
51     for pid, details in inventory.items():
52         if details["qty"] <= details["reorder"]:
53             print(f"{pid} | {details['name']} | Qty:{details['qty']} (Needs restock!)")
54
55 # Main Menu
56 while True:
57     print("\n===== INVENTORY MANAGEMENT SYSTEM =====")
58     print("1. Add Product")
59     print("2. Stock In")
60     print("3. Stock Out")
61     print("4. View Inventory")
62     print("5. Low Stock Alert")
63     print("6. Exit")
```

```

64
65     choice = input("Enter choice (1-6): ")
66
67     if choice == "1":
68         |     add_product()
69     elif choice == "2":
70         |     stock_in()
71     elif choice == "3":
72         |     stock_out()
73     elif choice == "4":
74         |     view_inventory()
75     elif choice == "5":
76         |     low_stock_alert()
77     elif choice == "6":
78         |     print("Exiting... Goodbye!")
79         |     break
80     else:
81         |     print(" Invalid choice! Please enter between 1-6.")
82

```

Output :

```

ers/Sarvesh/OneDrive/Desktop/code/project5.py

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Exit
Enter choice (1-6): 1
Enter Product ID: 897
Enter Product Name: Pen
Enter Price: 10
Enter Quantity: 50
Enter Reorder Level: 10
Product added successfully!

===== INVENTORY MANAGEMENT SYSTEM =====
1. Add Product
2. Stock In
3. Stock Out
4. View Inventory
5. Low Stock Alert
6. Exit
Enter choice (1-6): 4

===== INVENTORY =====
897 | Pen | Rs.10.0 | Qty:50 | Value: Rs.500.0

```